

PP3S Diary Week 2

Group E

May 2026

In the second week we refined our structure of the application and assigned the different components to pairs. Tekle and I are now responsible for the user module and the leadership module. We started by looking into the structure that had already been set up by other group members, so we could understand how our own modules should be built. Next, Tekle and I plan on to meet online to discuss how we want to structure our components. After that we will begin the implementation. (Ella)

I will work on the Users and Leaderboard part of the backend. First, I want to better understand the backend structure and improve my TypeScript skills. For the Users module, I plan to work on features like user registration, login, and handling user data. For the Leaderboard, I will focus on calculating user rankings based on portfolio performance and creating the necessary API endpoints. I also want to make sure the features work correctly with testing and mock data.(Tekle)

This week, I started working more seriously on the database design for our project. Based on our current system architecture, I created an initial schema draft for each backend module. At the moment, the database is divided into four main parts: authentication, portfolio, trade, and stock.

I also began thinking more carefully about the structure and consistency of the database. Besides defining the main tables and fields, I tried to organize the schemas in a modular way so they can be easier to maintain and expand in the future. Since this is still an early version, there are many details that still require further discussion and refinement.

Next week, I plan to continue improving the schemas and discuss the detailed requirements with my teammates during our next group meeting. After that, I want to further refine each schema, review the consistency between modules, and complete the missing relationships and foreign key connections. I also plan to audit all indexes to ensure that the database structure supports efficient queries and maintains good performance.

In addition, I would like to define enums more clearly and consistently in order to improve readability and reduce ambiguity in the system design. To help the rest of the team better understand the database architecture, I am

also planning to create an ER diagram and a data flow diagram to visualize the relationships between modules and the overall flow of the system. (Yifan)

As we discussed in the group meeting last week, Me and Mehmet worked on creating a basic example module so that we could get a deeper understanding of how we will be working throughout this project. We implemented GET and POST endpoints to simulate how data will be fetched and submitted in our application, and also added a basic logic calculation function to explore how business logic will be structured within our modules. This helped us align on the project structure, file organization, and how we will collaborate using our shared test branch on GitHub. (Doruk and Mehmet)

This week, we distributed implementations tasks based on the backend modules in a group meeting. To keep the code quality at a high standard and prepare for a production deployment, I created GitHub Workflows to lint for style errors in both backend and frontend. A small example test was also added to showcase the CI test phase in pull requests. I created a developer branch for staging the application onto the main (production) branch. I added Dockerfiles to the backend and created a compose file to locally deploy the Postgres database and backend for development. (David)

In the group meeting, I showcased our new group members the technology that we have decided to use and demonstrated the project structure. I gave a small tutorial on how the usual development cycle could be from using the Linear board, to creating a pull request to the dev branch. After the group meeting, I adjusted my developer environment and tested out all the other technologies that was recommended by the other group members, which I will be giving a report tomorrow, what could be optimally integrated with the backend work. (Moritz)

I added branch rules to the GitHub repository to prevent accidents such as accidentally pushing directly instead of creating a PR and requesting a code review. I helped developers during their first PRs deal with technical difficulties. I also ensured that all the diaries were written and organized the voting for the decisions about the next meeting. (David)

As discussed per last week, I added login and signup pages that directly interact with the auth routes provided by BetterAuth in the backend. For now there will be a default user. The real authentication will only be enabled at a later stage, once the backend is farther developed. I'm still testing before creating a PR into dev (Oliver)